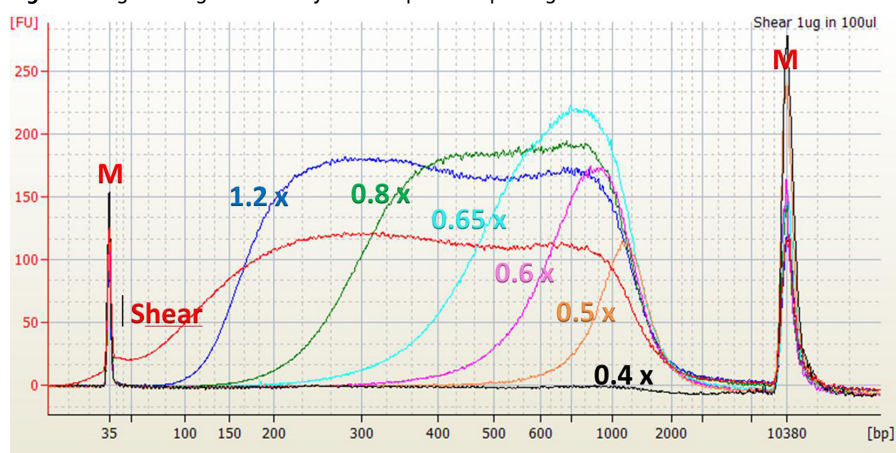


Left Side Size Selection

As a general rule, increasing the ratio of **SPRIselect** volume to sample volume will increase the efficiency of binding smaller fragments. **Figure 1** illustrates this relationship.

Figure 1 Agilent High Sensitivity DNA chip Electropherogram.



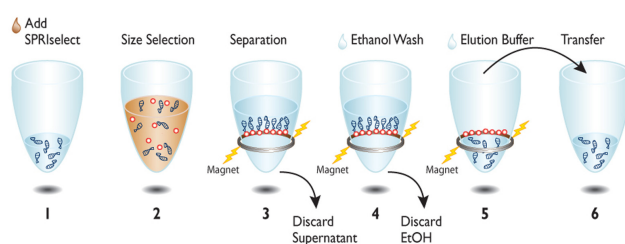
M = upper and lower markers for High Sensitivity DNA chip

Shear = 1 μ L of 20 ng/ μ L input control sample in water

1.2x to 0.4x = 1 μ L of shear, size selected with given ratio of **SPRIselect** volume to sample volume.

Figure 2

Left Side Size Selection Process Overview



1. Thoroughly shake the **SPRIselect** bottle to resuspend the **SPRI** beads. Following the trend depicted in **Figure 1**, add the required volume of **SPRIselect** for the desired ratio to the sample.

TIP Volume of sample * ratio = volume of **SPRIselect**.

Example: 50 μ L sample * 0.8x ratio = 40 μ L of **SPRIselect**

2. Mix the total reaction volume by pipetting 10 times and incubate at RT for 1 minute
OR

vortex for 1 minute at an appropriate speed until homogenous (depending on labware and total volume).

NOTE Insufficient mixing of sample and SPRIselect will lead to inconsistent size selection results. Make sure to mix well.

3. Place the reaction vessel on an appropriate magnetic stand or plate and allow the SPRI beads to settle to the magnet. Settle times will vary; a higher initial sample volume, higher **SPRIselect** ratio or weaker magnets will require a longer settle time. Remove and discard the clear supernatant.

NOTE Care should be taken not to aspirate more than a trace amount of beads during this step, as the desired library is associated with the beads. Significant bead loss will result in reduced yield.

4. With the reaction vessel still on the magnet, add 180 μ L of 85% ethanol (non-denatured) and incubate at RT for 30 seconds. Remove and discard the ethanol supernatant.

NOTE Care should be taken not to aspirate more than a trace amount of beads during this step, as the desired library is associated with the beads. Significant bead loss will result in reduced yield.

5. To elute the sample:

- a. Remove the reaction vessel from the magnet and add ≥ 20 μ L of molecular biology grade water or standard buffer solution such as Tris or TE.

NOTE Elution volume should be large enough so that the liquid level is high enough for the beads to settle to the magnet.

- b. Mix the total elution volume by pipetting 10 times to resuspend the beads and incubate at RT for 1 minute
OR
vortex for 1 minute at an appropriate speed until homogenous (depending on labware and total volume).
- c. Place the reaction vessel on an appropriate magnetic stand or plate and allow the SPRI beads to settle to the magnet. Settle times will vary; a higher elution volume or weaker magnets will require a longer settle time.

6. Transfer the eluate (size selected sample) to an appropriate storage vessel.